School of Mathematical Sciences

A Level 3 Module Spring 2024/25

Time Series Analysis

Instructions:

Time allowed: Three Hours

Answer ALL questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Only a calculator from approved list A may be used in this examination.

Basic Models	**Scientific Calculators**	
Aurora HC133	Aurora AX-582	Casio FX82 family
Casio HS-5D	Casio FX83 family	Casio FX85 family
Deli – DL1654	Casio FX350 family	Casio FX570 family
Sharp EL-233	Casio FX991 family	Sharp EL-531 family
	Texas Instruments TI-30 family	Texas BA II+ family

Permitted resources:
Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

Prohibited resources:
All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Appended material: None

Additional material: None

Information for Invigilators: None

Exam Papers must not be removed from the examination venue.

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2. You should show your working and/or justification clearly in all answers that involve calculations or derivations.

   (a) The process $\{X_t : t \in \mathbb{Z}\}$ is such that

   $$X_t = 1 + 2t + aX_{t-1} + Z_t$$

   where a is a real-valued, non-zero constant. Suppose that the process $\{Y_t : t \in \mathbb{Z}\}$ is defined:

   $$Y_t = \nabla^2 X_t$$

   where ∇ is the first difference operator. Show that

   $$Y_t = aY_{t-1} + Z_t - 2Z_{t-1} + Z_{t-2}.$$

   [2 marks]

   (b) Determine the set of values of a for which the process $\{Y_t : t \in \mathbb{Z}\}$ is (weakly) stationary.

   [3 marks]

   (c) Assuming that a is such that $\{Y_t : t \in \mathbb{Z}\}$ is (weakly) stationary, write this process as a moving average process of infinite order. You should write your final answer in its simplest form.

   [7 marks]

   (d) Suppose that $a = \frac{1}{2}$. Using your answer to (c), determine the autocovariance function of $\{Y_t : t \in \mathbb{Z}\}$ in terms of σ_z^2.

   [13 marks]

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2. You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The (weakly) stationary process $\{X_t : t \in \mathbb{Z}\}$ measures the number of celebration cake orders made at a bakery per year, where

$$X_t = 150 + 0.4X_{t-1} + Z_t - 0.5Z_{t-1}.$$

(a) Show that the mean of this process is 250.

[3 marks]

(b) Show that the process $\{X_t : t \in \mathbb{Z}\}$ is invertible and, hence, write this process in the form

$$Z_t = \sum_{j=0}^{\infty} b_j \left(X_{t-j} - a\right)$$

where the coefficients a and b_j $(j \in \{0, 1, 2, ...\})$ are real-valued constants.

[11 marks]

(c) Observed values of $\{X_t : t \in \mathbb{Z}\}$ for the years 2021-2024 are shown in the table below.

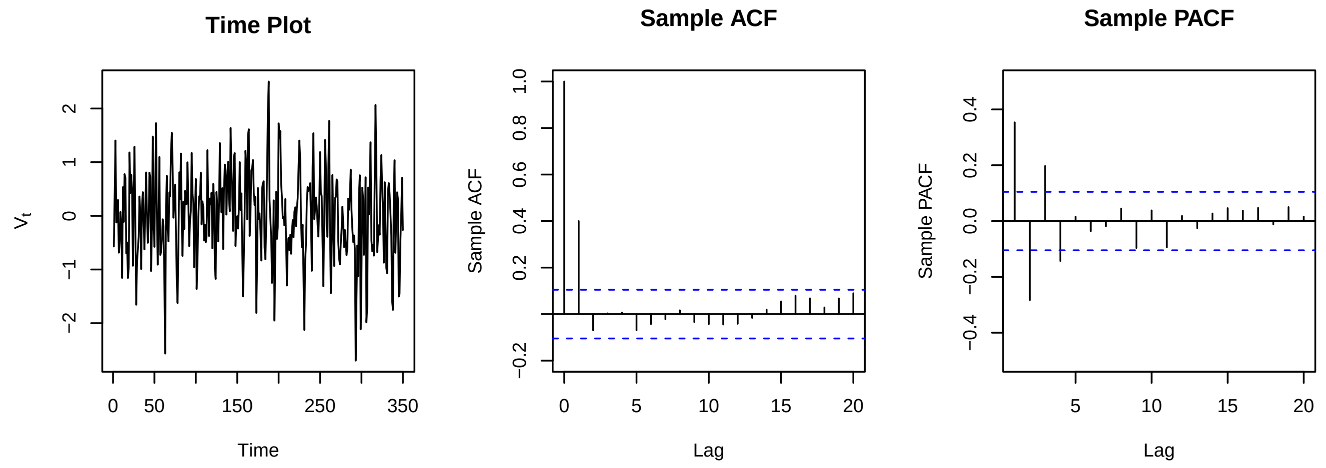
Year (t)	2021	2022	2023	2024
No. of orders (x_t)	240	268	254	272

Using this information, produce optimal predictions for the numbers of celebration cake orders for the years 2025 and 2026. Here, the term ‘optimal’ implies that the predicted values are those that minimise the mean square prediction error at corresponding time points.

[11 marks]

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2. You should show your working and/or justification clearly in all answers that involve calculations or derivations.

(a) A dataset contains 350 sequentially sampled values of the process $\{V_t : t \in \mathbb{Z}\}$. The figure below shows a time plot, a plot of the sample autocorrelation function (ACF) and sample partial autocorrelation function (PACF) against the lag, for this sample.



Using the plots above, suggest an appropriate model for the process $\{V_t : t \in \mathbb{Z}\}$. You should justify your answer and provide approximate estimates of model parameters.

[14 marks]

(b) A statistician fits the following model to data from a process $\{X_t : t \in \mathbb{Z}\}$:

$$X_t = 0.4X_{t-1} + Z_t - 0.5Z_{t-1} + 0.06Z_{t-2}.$$

The model residuals, $\{Z'_t : t \in \mathbb{Z}\}$, are examined and appear to be an $AR(1)$ process of the form

$$Z'_t = 0.5Z'_{t-1} + Z_t.$$

Identify the type of model that the statistician should fit next, including its order.

[6 marks]

(c) The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \frac{1}{4}Y_{t-1} + \frac{1}{8}Y_{t-2} + Z_t,$$

and the process $\{U_t : t \in \mathbb{Z}\}$ is defined

$$U_t = Y_t + \frac{2}{3}Y_{t-1} + \frac{1}{24}Y_{t-2}.$$

Determine the equation for U_t in terms of $\{Z_t : t \in \mathbb{Z}\}$ and past terms of $\{U_t : t \in \mathbb{Z}\}$. What type of process is $\{U_t : t \in \mathbb{Z}\}$?

[5 marks]

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2. You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = -\frac{1}{2}X_{t-1} + \frac{3}{16}X_{t-2} + Z_t - \frac{1}{4}Z_{t-1}$$

(a) Determine the spectral density function of $\{X_t : t \in \mathbb{Z}\}$, expressing your answer in its simplest form as a function of $\cos w$ (where $w \in (-\pi, \pi]$).

[5 marks]

(b) Sketch the spectral density function from (a) for values of w where $w \in [0, \pi]$. Explain what this sketch suggests regarding the likely behaviour of the process $\{X_t : t \in \mathbb{Z}\}$ over time.

[6 marks]

(c) A linear filter is applied to $\{X_t : t \in \mathbb{Z}\}$ to create a new process $\{Y_t : t \in \mathbb{Z}\}$, where

$$Y_t = X_t - \frac{9}{16}X_{t-2}$$

Determine the squared gain with respect to this linear filter.

[5 marks]

(d) Using your answers to (a) and (c), determine the spectral density function of $\{Y_t : t \in \mathbb{Z}\}$.

[3 marks]

(e) Sketch the spectral density function from (d) and explain briefly how the behaviour of $\{Y_t : t \in \mathbb{Z}\}$ over time differs from that of $\{X_t : t \in \mathbb{Z}\}$.

[6 marks]

End